

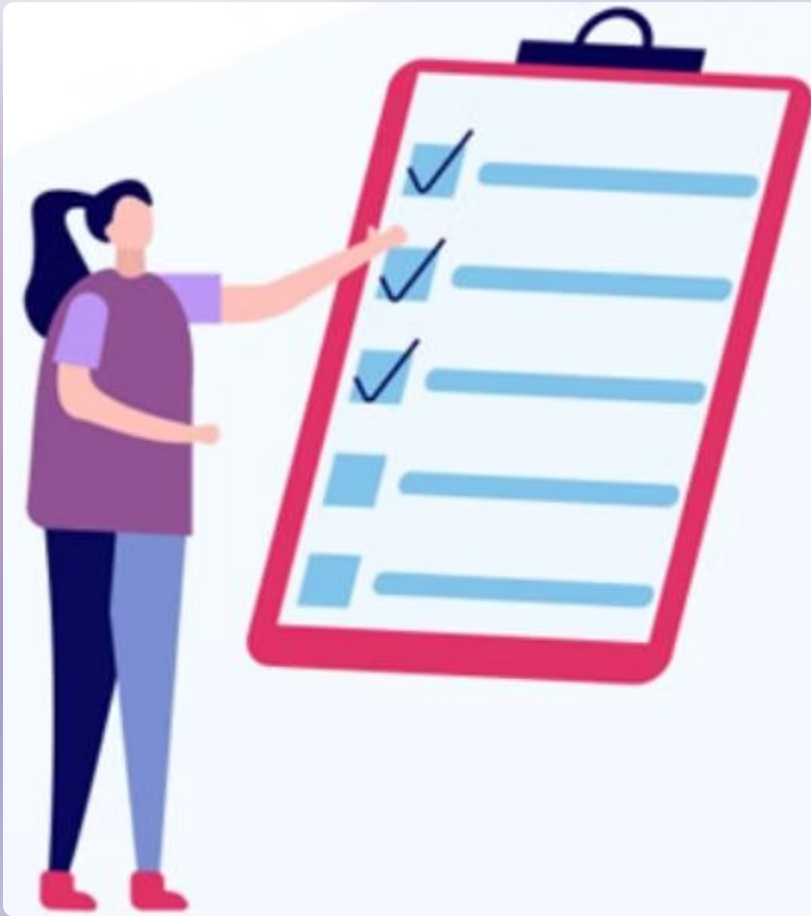


Machine Learning

A beginner-friendly introduction to how computers learn from data

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Learning Objectives

By the end of this lesson, you will be able to:

- 1 Understand AI, ML & Deep Learning
Analyze the distinctions and real-world applications of machine learning, deep learning, and artificial intelligence.
- 2 Explore ML Models
Differentiate among various machine learning models and explore how each learns from data to predict outcomes.
- 3 Python Libraries
Explore Python libraries for data manipulation, visualization, and implementation of machine learning algorithms.



What Is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that helps systems **learn and improve automatically from experience** — without being explicitly programmed.

Arthur Samuel coined the term **machine learning** in **1959**. It enables programs to learn automatically, making computers more intelligent without human intervention.

Traditional Approach

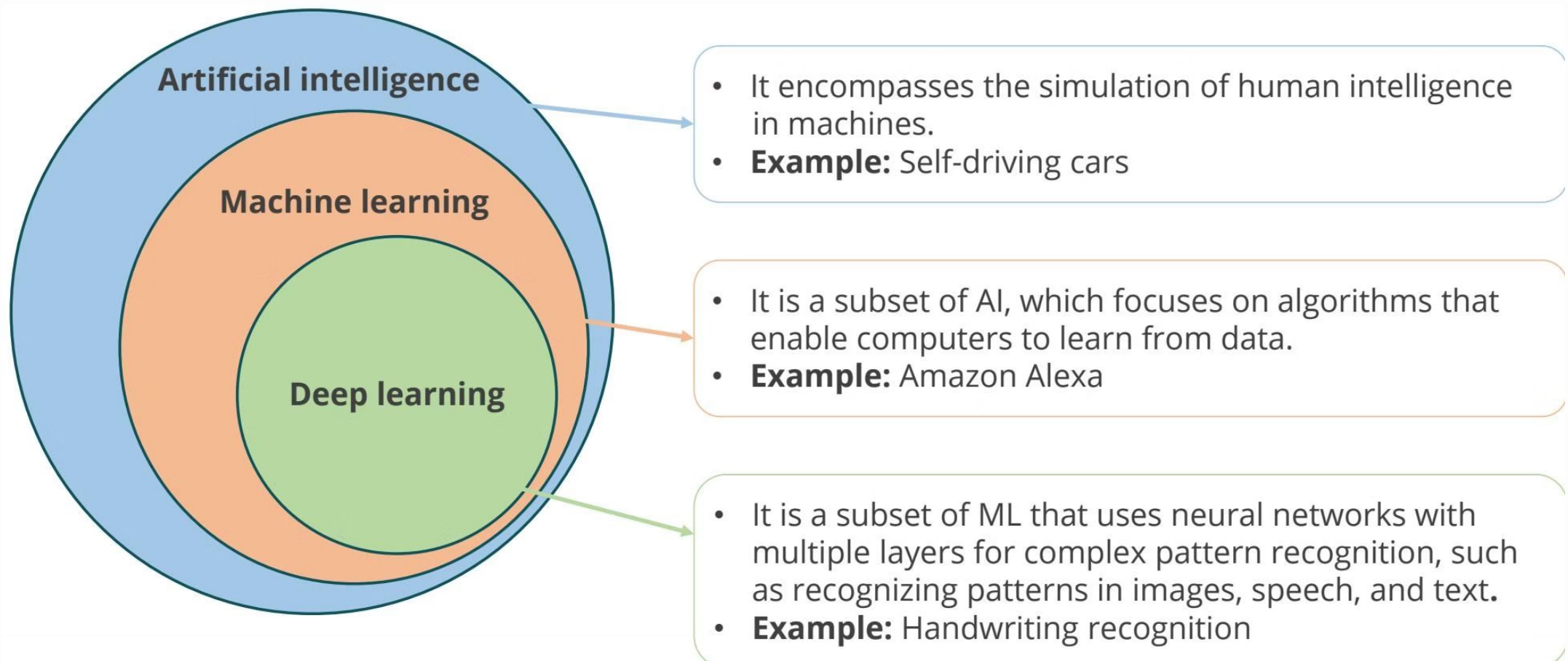
Uses predefined rules written by humans. Requires manual feature engineering. Struggles with complex, unstructured data.

Machine Learning Approach

Learns patterns from data automatically. Handles images, text, and audio. Performance improves with more data and iterations.

AI vs. ML vs. Deep Learning

These three terms are often used interchangeably – but they are different. Think of them as **nested layers**, each more specialized than the last.



Artificial Intelligence

Simulates human intelligence in machines.

Example: Self-driving cars



Machine Learning

A subset of AI – algorithms that learn from data.

Example: Amazon Alexa



Deep Learning

A subset of ML using neural networks for complex pattern recognition.

Example: Handwriting recognition

Applications & Algorithms of ML

Real-World Applications



Spam Filtering

Detects and blocks unwanted emails automatically.



Social Media Analysis

Analyzes trends, sentiments, and user behavior.



Customer Service Chatbots

Provides instant, automated customer support.



Recommendation Systems

Suggests products, movies, or songs based on preferences.

What ML Algorithms Do



Predict Outcomes

Forecast future results from historical data.



Classify Features

Sort data into categories automatically.



Improve Performance

Get better with more data and iterations.



High-quality data is crucial – it ensures better predictions and insights.

Types of Machine Learning

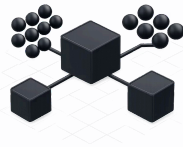
ML is divided into **four main categories**, each with a unique way of learning from data.



Supervised Learning

Uses **labeled data** — both inputs and outputs are known. Algorithms: Linear Regression, Decision Trees, SVM.

Example: Spam filtering, predicting crop yield.



Unsupervised Learning

Finds **hidden patterns** in unlabeled data without guidance.

Example: Image segmentation, anomaly detection, user grouping.



Semi-Supervised Learning

Uses a **small amount of labeled** + large amount of unlabeled data. Falls between supervised and unsupervised.

Example: Google Photos face recognition.



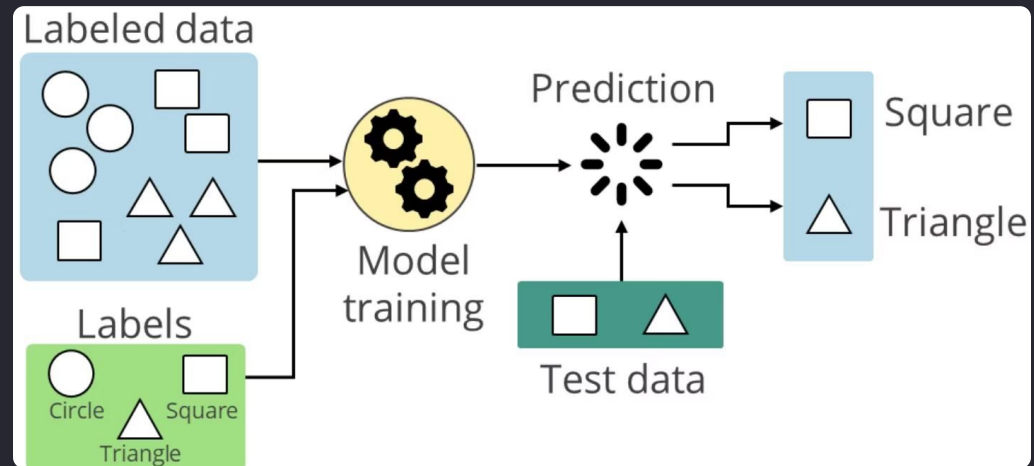
Reinforcement Learning

Learns by **trial and error** — rewarded for correct actions, penalized for wrong ones.

Example: Self-driving cars, YouTube recommendations.

Supervised vs. Unsupervised Learning

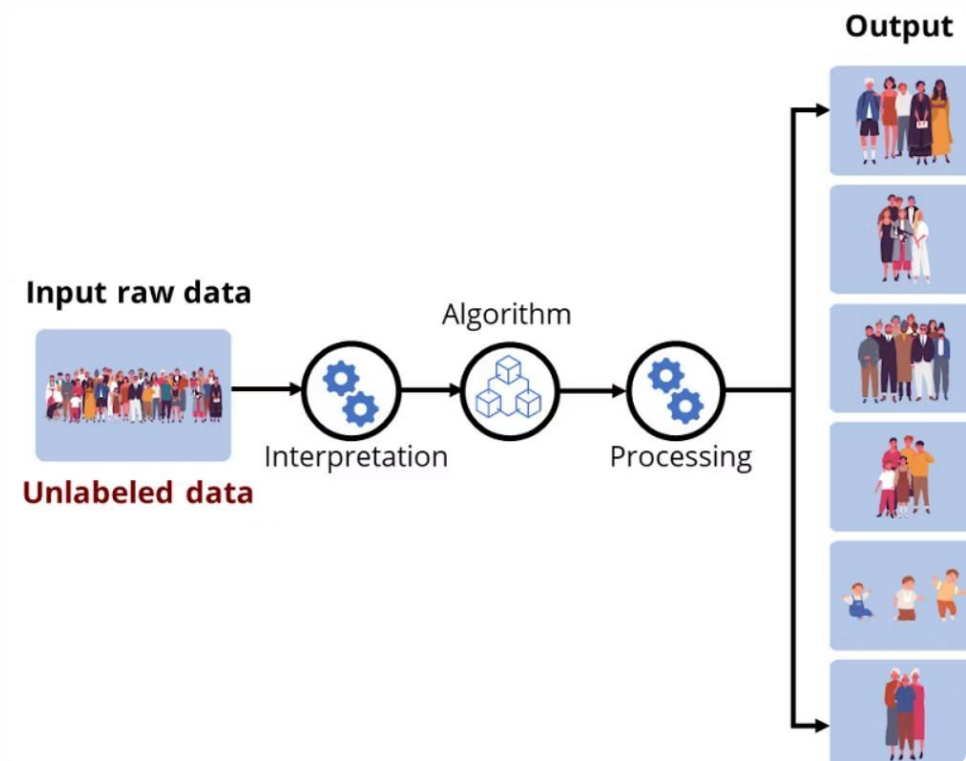
Supervised Learning



Labeled data → Model Training → Prediction. Both inputs and outputs are known. The model learns the mapping between them.

- Linear Regression
- Decision Trees
- Logistic Regression
- Support Vector Machines

Unsupervised Learning



Unlabeled data → Algorithm → Discovers hidden groups. No predefined outputs – the model finds patterns on its own.

- Image segmentation for object detection
- User group identification
- Anomaly detection over landscapes

Reinforcement Learning & Semi-Supervised Learning

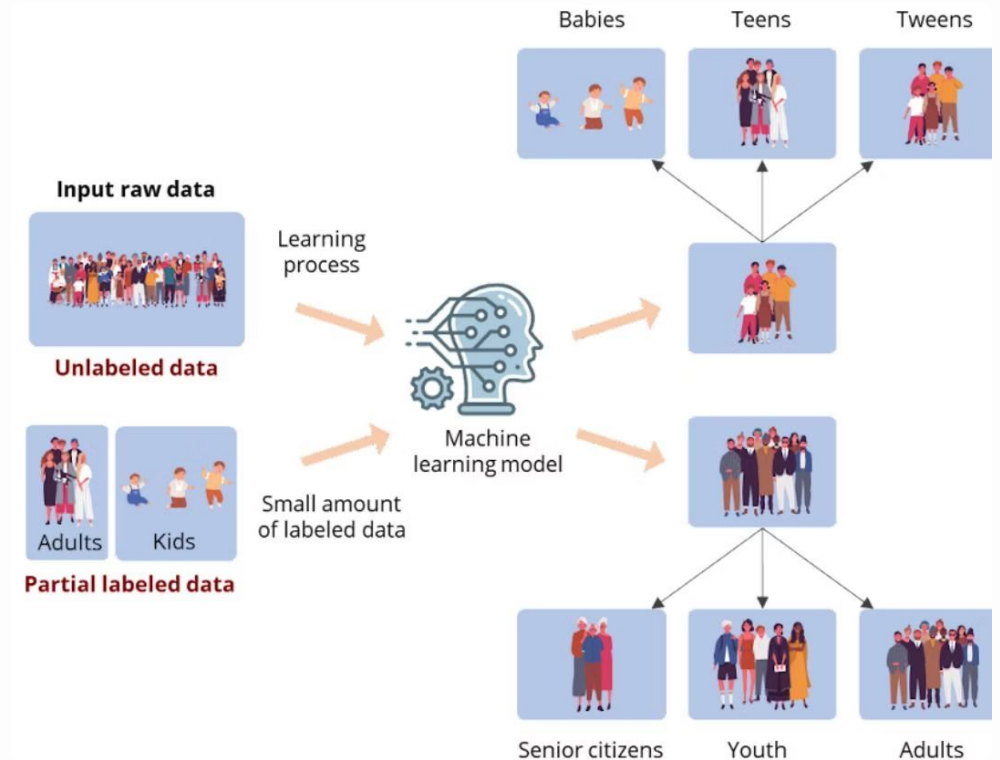
Reinforcement Learning



An **agent** interacts with an environment, receives **rewards** for correct actions and **penalties** for wrong ones — improving over time.

Examples: YouTube recommendations, self-driving cars, game-playing bots, autocorrect tools, search engines.

Semi-Supervised Learning



Uses a **small amount of labeled data** combined with a large amount of unlabeled data. Falls between supervised and unsupervised learning.

Example: Google Photos — labels a few images, then identifies others by analyzing visual features like shapes and colors.

Python Libraries for Machine Learning

Python is the most popular language for ML. These libraries make it easy to work with data and build models:

1

NumPy

Powerful library for **numerical computing** in Python. Handles arrays and mathematical operations efficiently.



Matplotlib

Used for **data visualization** and graphical plotting – creates charts, graphs, and figures.



Pandas

Versatile **data manipulation** library offering powerful data structures like DataFrames.

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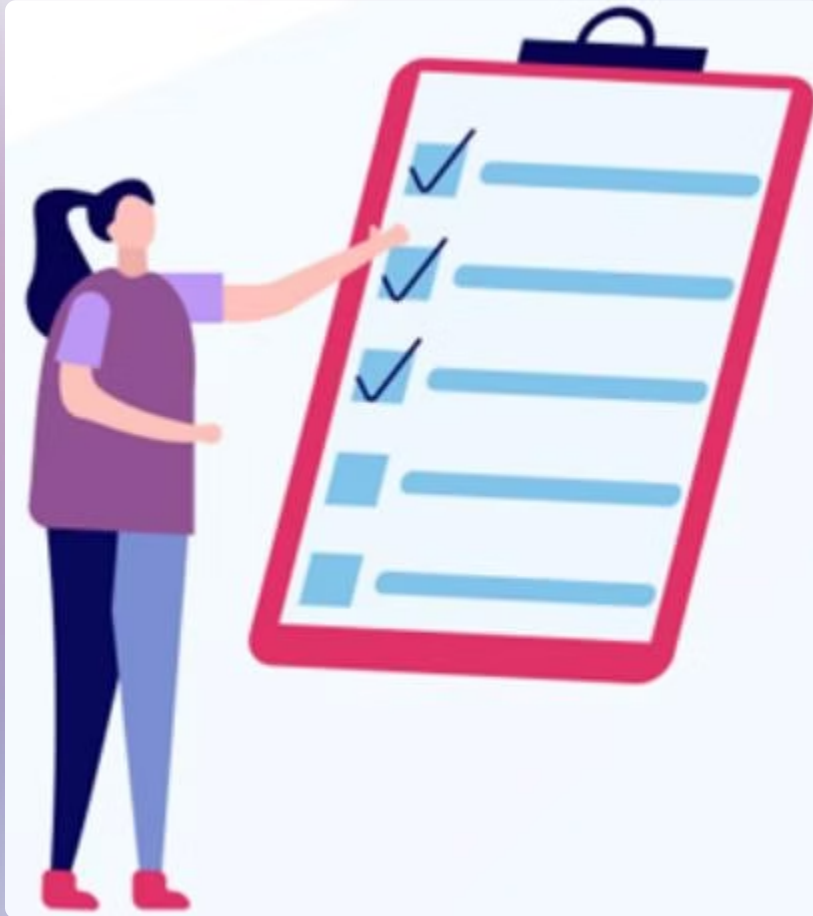
SciPy

Solves **mathematical equations** and processes scientific algorithms.



Scikit-learn

Offers **efficient ML algorithms** – the go-to library for building and evaluating machine learning models.



Key Takeaways

1 Machine Learning = Learning from Data

ML refers to a machine's ability to learn from data and replicate human behavior – without being explicitly programmed.

2 $AI \supset ML \supset \text{Deep Learning}$

AI includes ML and Deep Learning, each with unique capabilities for simulating intelligence at different levels of complexity.

3 Four Types of ML

Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning – each learns differently from data.

4 Python Powers ML

Libraries like NumPy, Pandas, Matplotlib, SciPy, and Scikit-learn make it easy to build and deploy ML models efficiently.

Thank You!